LUMAFLOW · TRACEABILITY MATRIX

LumaFlow traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.3	2026-10-25	Amish Chadha	Generated by .kit/trace.py
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LumaFlow traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 18 user requirements, 19 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-002	PR-001	Drinking-water flow	LMF-CAL-001, LMF-BLD-001, BOM	Analysis (CAL A3), TRL 3; restrictor rating	Met by design [A3]
UR-001	PR-002	Class A dose level in clear water	LMF-CAL-001, LMF-STM-001	Analysis (ray trace, CAL B5), TRL 3; biodosimetry at TRL 4	At risk: 50.9 (laminar) to 74.9 (plug) [B5]; 47.3 with the 275 nm response factor of 0.93 [B10], margin 18 % against the 20 % rule; 37.5 with a worst-case LED derating stack [B11] (STM S33); mixing to verify (S34)
UR-003	PR-003	Dose at the test water quality	LMF-CAL-001, LMF-STM-001	Analysis (CAL B6), TRL 3	Not met: 17.0 to 21.3 [B6], 15.8 with the 275 nm factor [B10]; kept visible by decision D1
UR-004, UR-003	PR-004	Dose monitor and fail-safe	LMF-CAL-001, LMF-BLD-001, LMF-STM-001, LMF-OPT-001	Analysis (CAL C1 to C5), TRL 3; firmware review and test at TRL 4	At risk: 13.4 nA at 90 %/cm; alarm at S/S_ref 0.85, set from the 275 nm weighted RED, below about 89 %/cm and on one failed LED [C1, C3, C4]; mapping and 1 s fail-safe need firmware and a test
UR-005	PR-005	Switch with the flow	LMF-CAL-001, LMF-BLD-001, BOM	Analysis (CAL D2, D5), TRL 3	Met on paper: the outlet pressure sensor wakes the unit when the faucet opens, LEDs on before the valve opens; 0.24 s to confirm flow at 0.3 L/min [D2, D5]
UR-005, UR-006	PR-006	No mercury, no warm-up	LMF-CAL-001	Inspection of BOM and LED datasheet, TRL 3	Met by design
UR-007	PR-007	Window under pressure	LMF-CAL-001, LMF-BLD-001	Analysis (CAL G4), TRL 3; pressure test at TRL 4	Met on paper: 6.18 [G4]
UR-008	PR-008	Pressure drop	LMF-CAL-001, LMF-BLD-001, LMF-STM-001	Analysis (CAL G2), TRL 3; datasheet Kv at part selection	At risk: 0.15 on assumed Kv [G2]; 0.59 with a direct-acting valve of Kv 0.10 [G9] (STM S22)
UR-009	PR-009	Energy use	LMF-CAL-001, LMF-BLD-001, LMF-STM-001, LMF-OPT-001	Analysis (CAL E1, E2), TRL 3	Met on paper: 22.6 and 0.23 [E1, E2]
UR-010	PR-010	LED board temperature	LMF-CAL-001, LMF-BLD-001, BOM	Analysis (CAL F3), TRL 3	Met on paper: 37.3 to 45.1 [F3]; the film coefficient in the cap bore is about 490 W/(m² K) for developing flow, above the 150 low end (STM S35)

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-011	PR-011	Food-safe wetted parts	LMF-CAL-001, LMF-BLD-001, LMF-STM-001	Inspection of the model and BOM, TRL 3; PTFE transmittance at TRL 4	Met by design: window gasket and sensor washer now expanded PTFE (LMF-DEC-001 D-S1); light reaching the acetal cap through 3 mm of PTFE to verify at TRL 4 (STM S10)
UR-012	PR-012	Contain the UV-C	LMF-CAL-001, LMF-BLD-001	Inspection (design review, model check), TRL 3	Met by design: lid reed switch; interlock loop in the LED head plug on a cable 39 mm short of letting the head clear the cap
UR-013	PR-013	Electrical safety	LMF-CAL-001	Inspection, TRL 3	Met by design
UR-014	PR-014	Fit under a sink	LMF-CAL-001, LMF-BLD-001, LMF-OPT-001	Analysis (model, CAL H1), TRL 3	Met on paper: 277 x 100 x 322 [H1]
UR-015, UR-017	PR-015	Easy service	LMF-CAL-001, LMF-BLD-001	Demonstration on a build, TRL 4	To verify at TRL 4
UR-016	PR-016	Cost and buildability	LMF-CAL-001, LMF-OPT-001, BOM	Analysis (priced BOM, CAL I1), TRL 3	Over target: USD 433, USD 93 over [I1]
UR-007	PR-017	Pressure protection (installation)	LMF-CAL-001, LMF-BLD-001, LMF-STM-001	Analysis (CAL G5), TRL 3; installation instructions	Met on paper: 3.09 MPa at 4 bar, 6.18 MPa at twice the setting [G5]
UR-017	PR-018	UV level at a glance	LMF-CAL-001, LMF-BLD-001, BOM	Analysis (CAL C6), TRL 3; bench check at TRL 4	Met by design: 2 of 5 at 90 %/cm with the weighted threshold, last segment out at the alarm [C6]
UR-018	PR-019	Honest claims	LMF-STM-001	Inspection of labels and documents, TRL 3	Met by design (BOM line 19; PRC safety; STM S5)

Gaps

None.